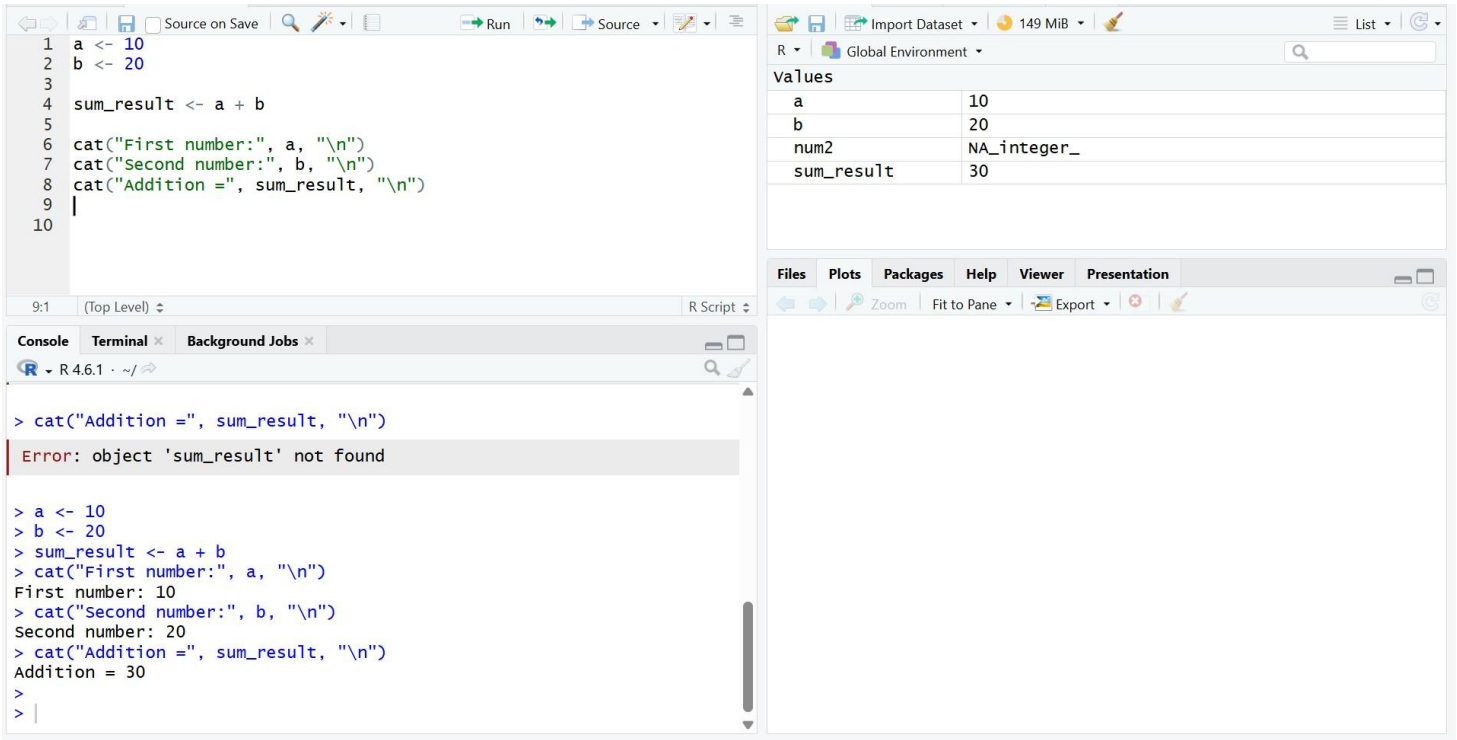
# Experiment-1



```
a <- 10
b <- 20

sum_result <- a + b

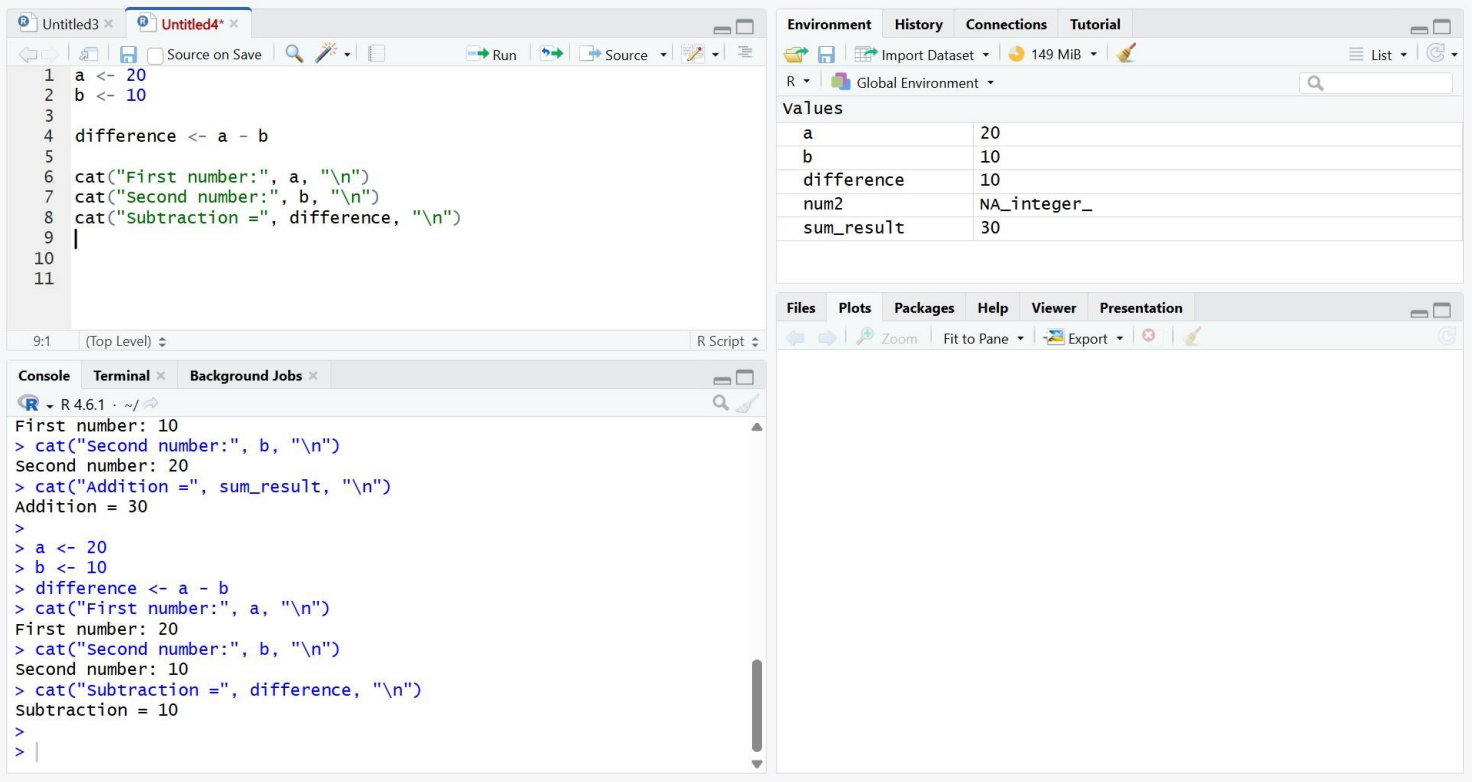
cat("First number:", a, "\n")
cat("Second number:", b, "\n")
cat("Addition =", sum_result, "\n")
```

```
> cat("Addition =", sum_result, "\n")
Error: object 'sum_result' not found

> a <- 10
> b <- 20
> sum_result <- a + b
> cat("First number:", a, "\n")
First number: 10
> cat("Second number:", b, "\n")
Second number: 20
> cat("Addition =", sum_result, "\n")
Addition = 30
>
>
```

| Values | |
|---|---|
| a | 10 |
| b | 20 |
| num2 | NA_integer_ |
| sum_result | 30 |

# Experiment-2



```
a <- 20
b <- 10

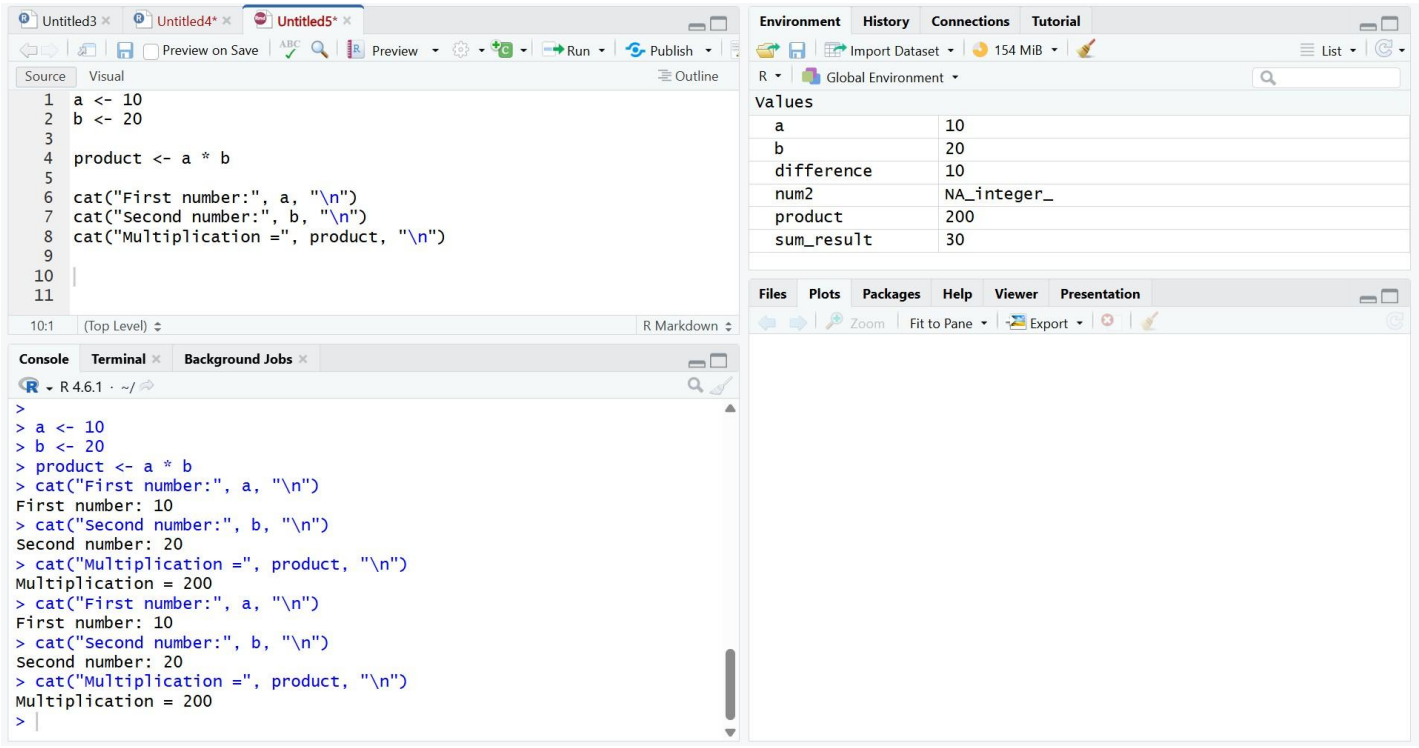
difference <- a - b

cat("First number:", a, "\n")
cat("Second number:", b, "\n")
cat("Subtraction =", difference, "\n")
```

```
First number: 10
> cat("Second number:", b, "\n")
Second number: 20
> cat("Addition =", sum_result, "\n")
Addition = 30
>
> a <- 20
> b <- 10
> difference <- a - b
> cat("First number:", a, "\n")
First number: 20
> cat("Second number:", b, "\n")
Second number: 10
> cat("Subtraction =", difference, "\n")
Subtraction = 10
>
>
```

| Values | |
|---|---|
| a | 20 |
| b | 10 |
| difference | 10 |
| num2 | NA_integer_ |
| sum_result | 30 |

# Experiment-3



```
a <- 10
b <- 20

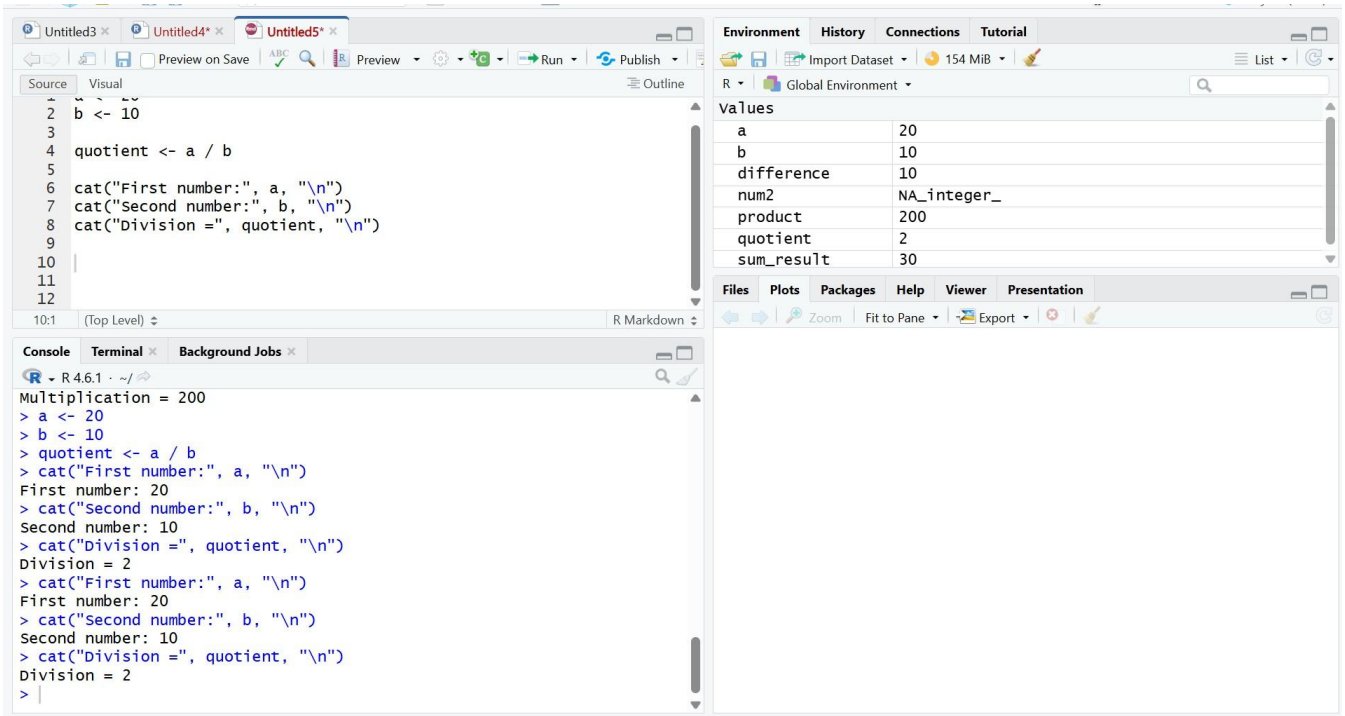
product <- a * b

cat("First number:", a, "\n")
cat("Second number:", b, "\n")
cat("Multiplication =", product, "\n")
```

| Values | |
|---|---|
| a | 10 |
| b | 20 |
| difference | 10 |
| num2 | NA_integer_ |
| product | 200 |
| sum_result | 30 |

```
> a <- 10
> b <- 20
> product <- a * b
> cat("First number:", a, "\n")
First number: 10
> cat("Second number:", b, "\n")
Second number: 20
> cat("Multiplication =", product, "\n")
Multiplication = 200
> cat("First number:", a, "\n")
First number: 10
> cat("Second number:", b, "\n")
Second number: 20
> cat("Multiplication =", product, "\n")
Multiplication = 200
>
```

# Experiment-4



```
b <- 10

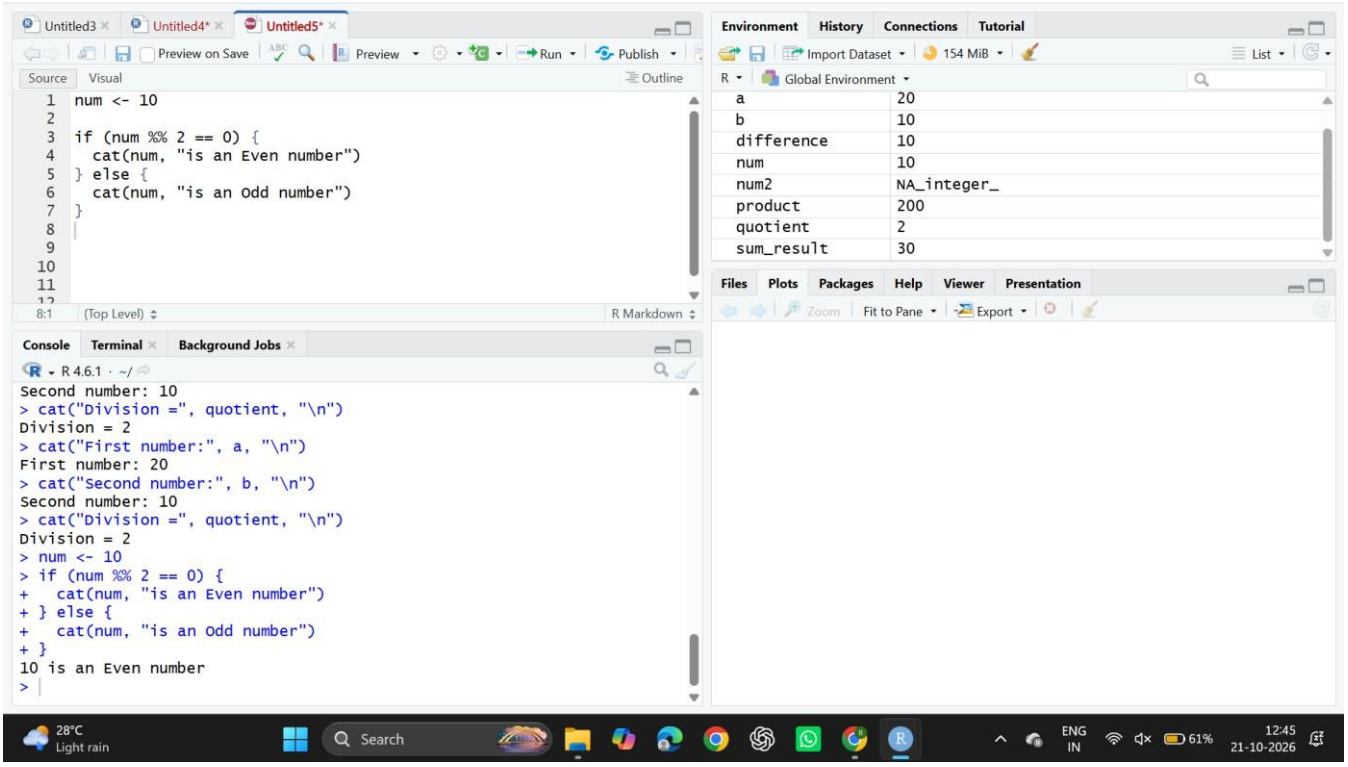
quotient <- a / b

cat("First number:", a, "\n")
cat("Second number:", b, "\n")
cat("Division =", quotient, "\n")
```

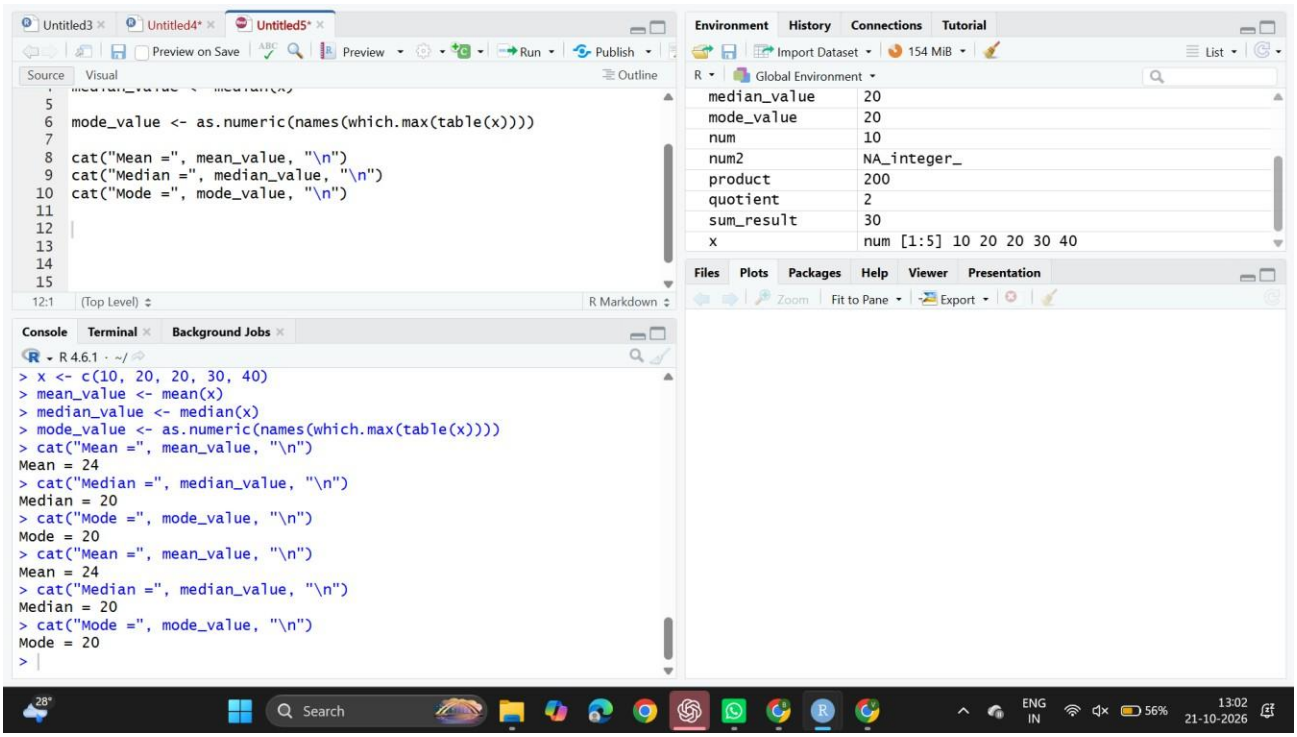
| Values | |
|---|---|
| a | 20 |
| b | 10 |
| difference | 10 |
| num2 | NA_integer_ |
| product | 200 |
| quotient | 2 |
| sum_result | 30 |

```
Multiplication = 200
> a <- 20
> b <- 10
> quotient <- a / b
> cat("First number:", a, "\n")
First number: 20
> cat("Second number:", b, "\n")
Second number: 10
> cat("Division =", quotient, "\n")
Division = 2
> cat("First number:", a, "\n")
First number: 20
> cat("Second number:", b, "\n")
Second number: 10
> cat("Division =", quotient, "\n")
Division = 2
>
```

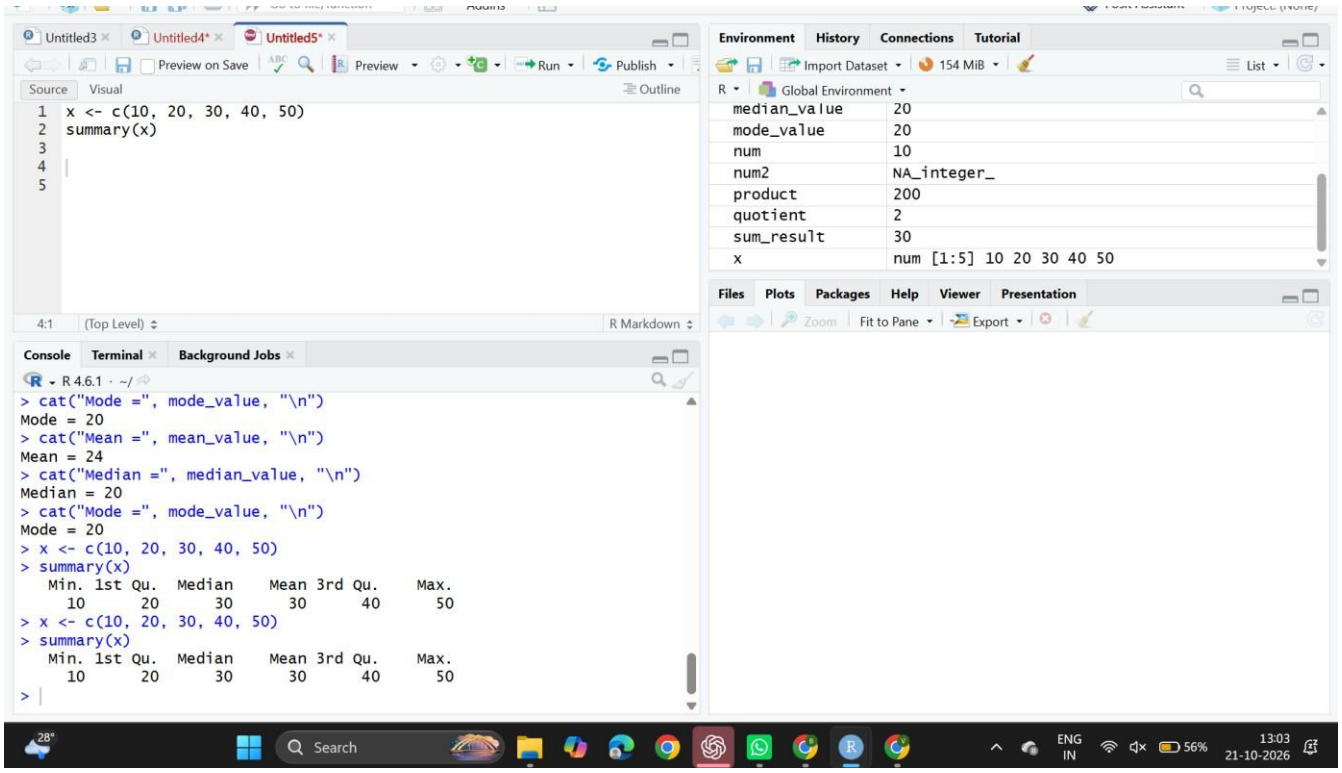
# Expeiment-5



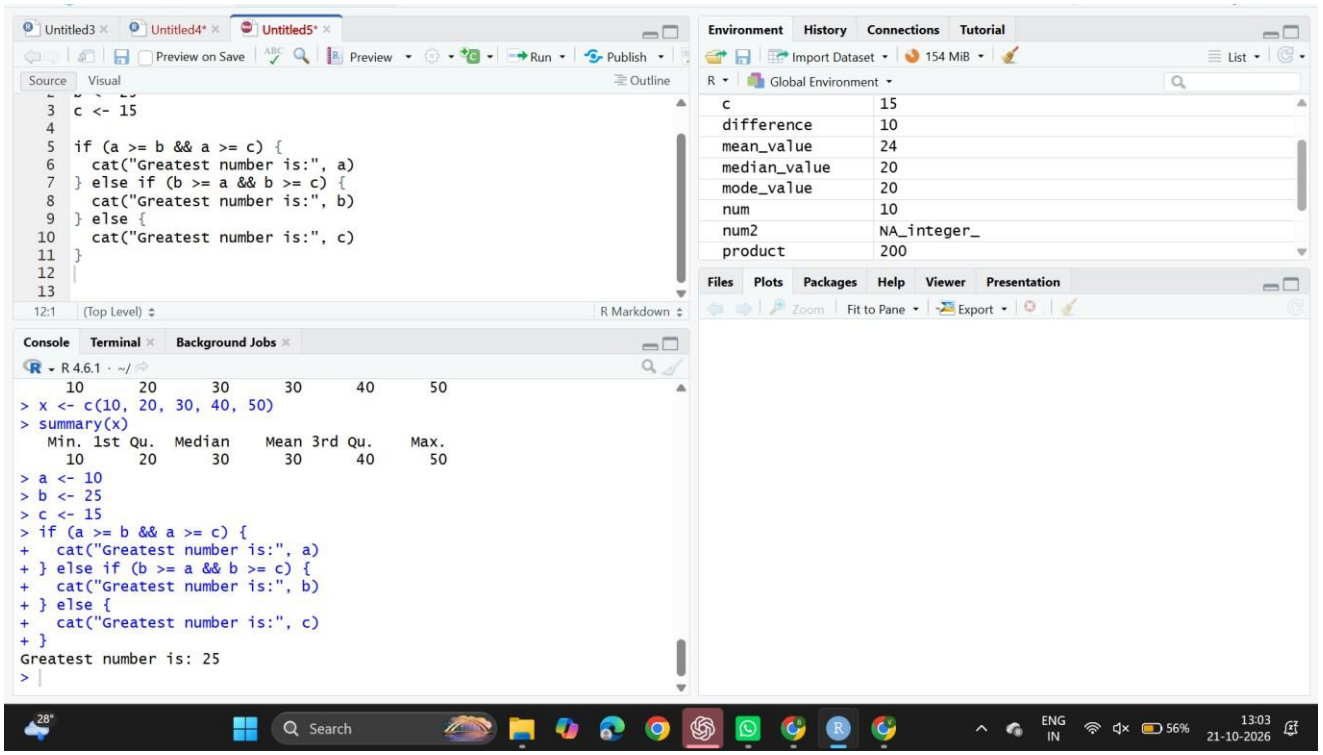

# Experiment-6



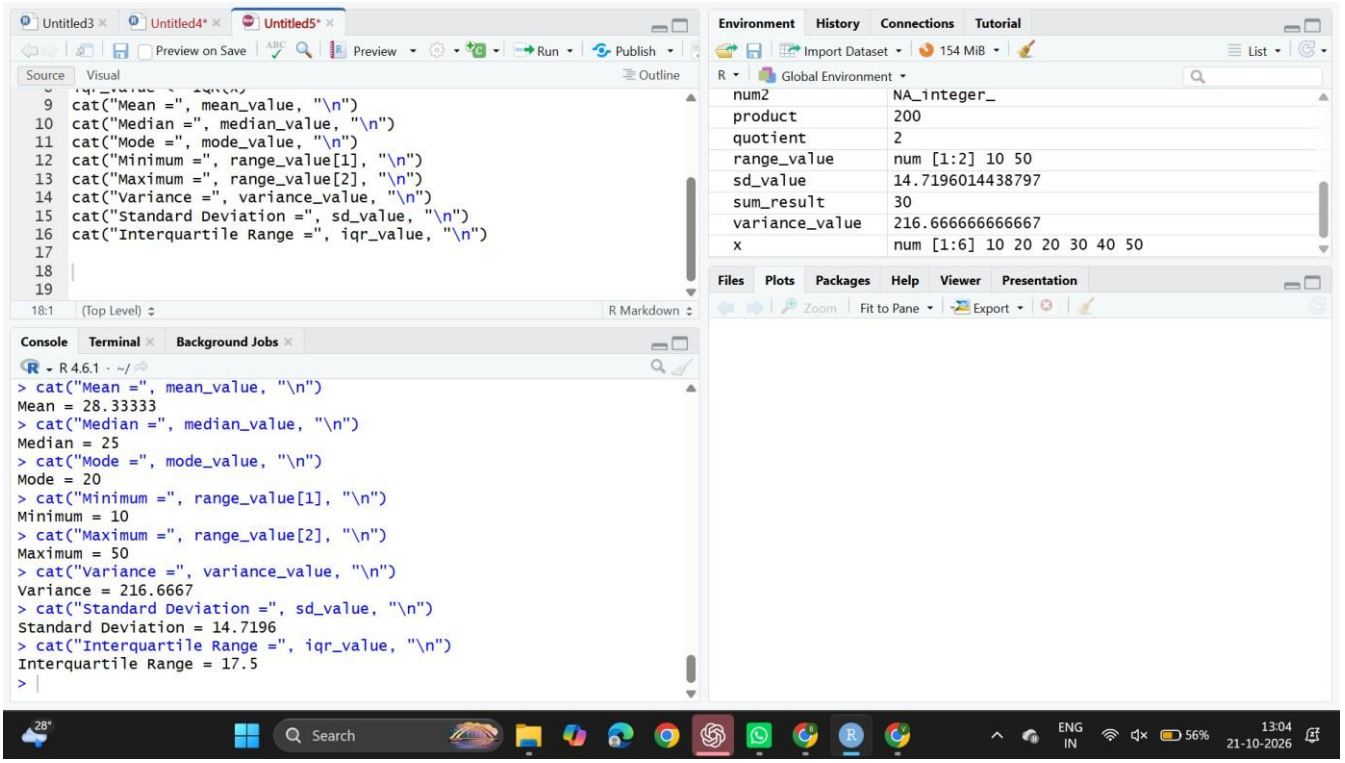
# Experiment-7




# Experiment-8



# Experiment-9

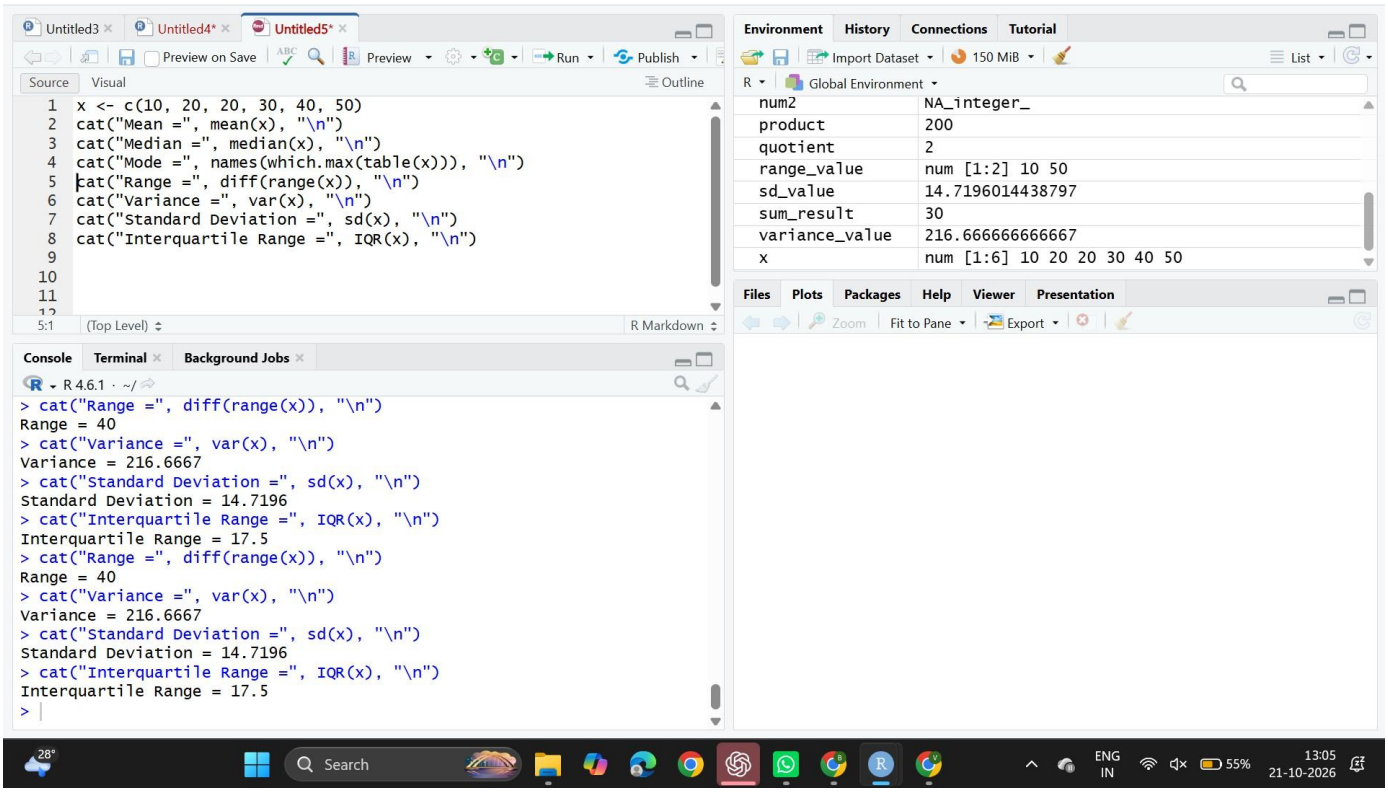


```
cat("Mean =", mean_value, "\n")
cat("Median =", median_value, "\n")
cat("Mode =", mode_value, "\n")
cat("Minimum =", range_value[1], "\n")
cat("Maximum =", range_value[2], "\n")
cat("Variance =", variance_value, "\n")
cat("Standard Deviation =", sd_value, "\n")
cat("Interquartile Range =", iqr_value, "\n")
```

```
> cat("Mean =", mean_value, "\n")
Mean = 28.33333
> cat("Median =", median_value, "\n")
Median = 25
> cat("Mode =", mode_value, "\n")
Mode = 20
> cat("Minimum =", range_value[1], "\n")
Minimum = 10
> cat("Maximum =", range_value[2], "\n")
Maximum = 50
> cat("Variance =", variance_value, "\n")
Variance = 216.6667
> cat("Standard Deviation =", sd_value, "\n")
Standard Deviation = 14.7196
> cat("Interquartile Range =", iqr_value, "\n")
Interquartile Range = 17.5
>
```

| | |
|---|---|
| num2 | NA_integer_ |
| product | 200 |
| quotient | 2 |
| range_value | num [1:2] 10 50 |
| sd_value | 14.7196014438797 |
| sum_result | 30 |
| variance_value | 216.666666666667 |
| x | num [1:6] 10 20 20 30 40 50 |

# Experiment-10

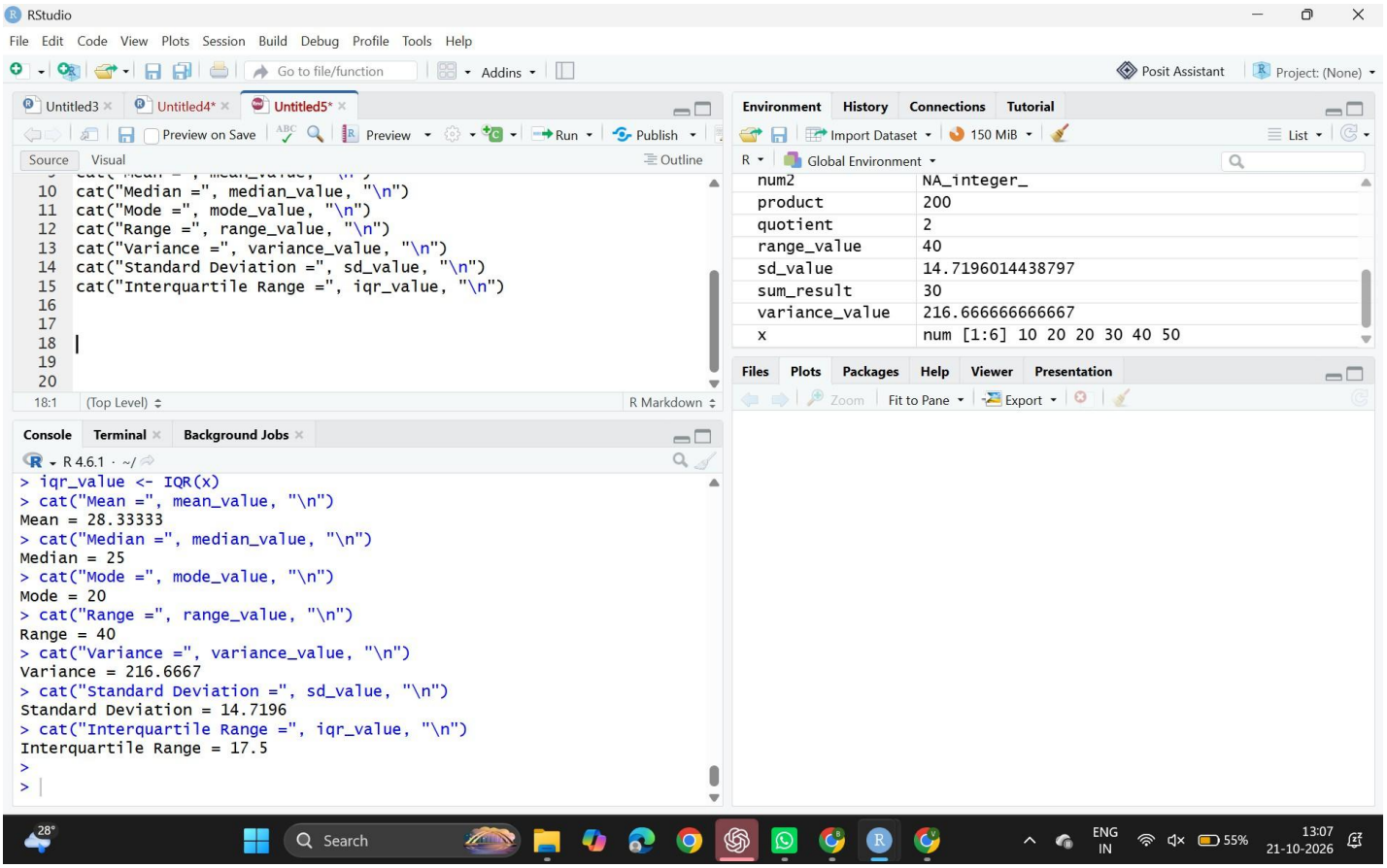


```
x <- c(10, 20, 20, 30, 40, 50)
cat("Mean =", mean(x), "\n")
cat("Median =", median(x), "\n")
cat("Mode =", names(which.max(table(x))), "\n")
cat("Range =", diff(range(x)), "\n")
cat("Variance =", var(x), "\n")
cat("Standard Deviation =", sd(x), "\n")
cat("Interquartile Range =", IQR(x), "\n")
```

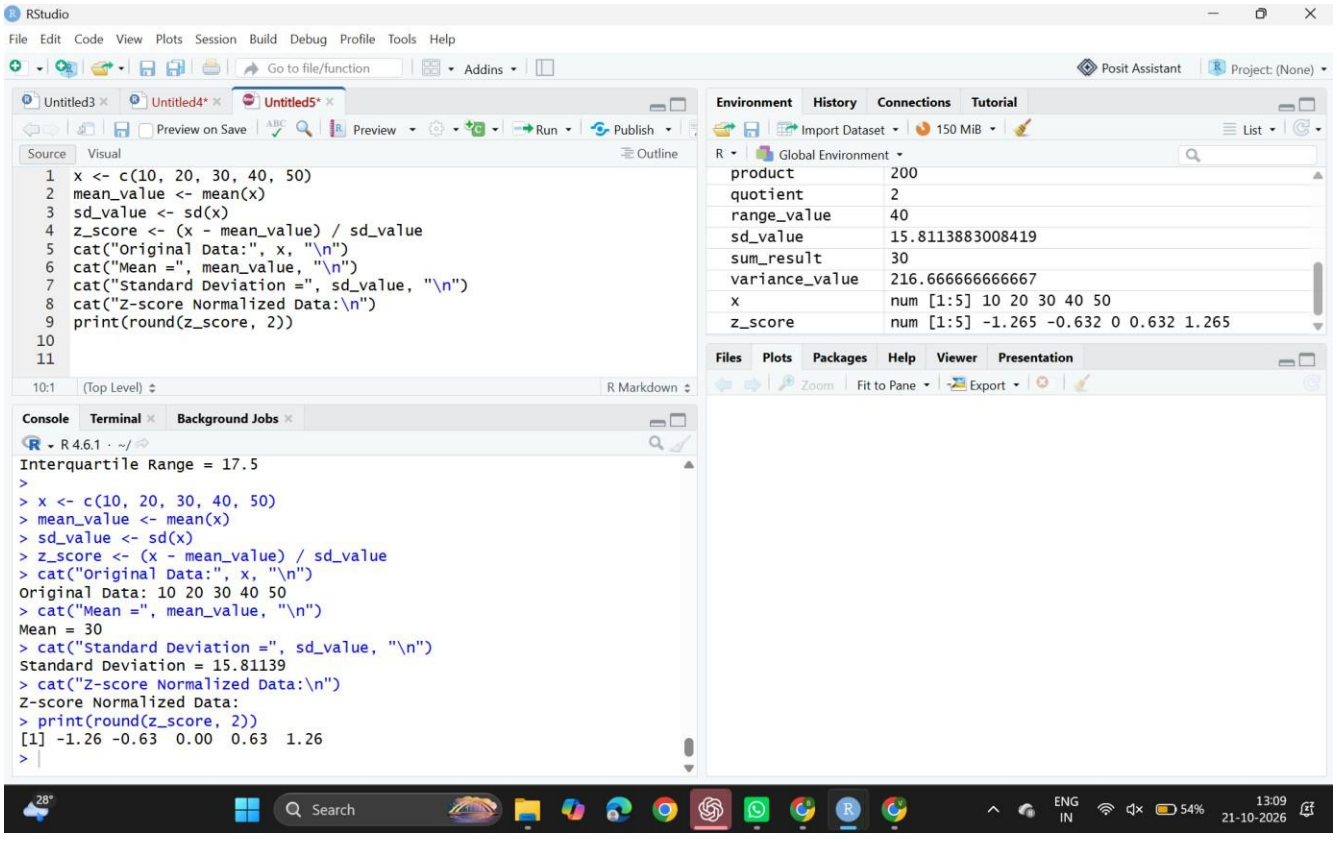
```
> cat("Range =", diff(range(x)), "\n")
Range = 40
> cat("Variance =", var(x), "\n")
Variance = 216.6667
> cat("Standard Deviation =", sd(x), "\n")
Standard Deviation = 14.7196
> cat("Interquartile Range =", IQR(x), "\n")
Interquartile Range = 17.5
> cat("Range =", diff(range(x)), "\n")
Range = 40
> cat("Variance =", var(x), "\n")
Variance = 216.6667
> cat("Standard Deviation =", sd(x), "\n")
Standard Deviation = 14.7196
> cat("Interquartile Range =", IQR(x), "\n")
Interquartile Range = 17.5
>
```

| | |
|---|---|
| num2 | NA_integer_ |
| product | 200 |
| quotient | 2 |
| range_value | num [1:2] 10 50 |
| sd_value | 14.7196014438797 |
| sum_result | 30 |
| variance_value | 216.666666666667 |
| x | num [1:6] 10 20 20 30 40 50 |

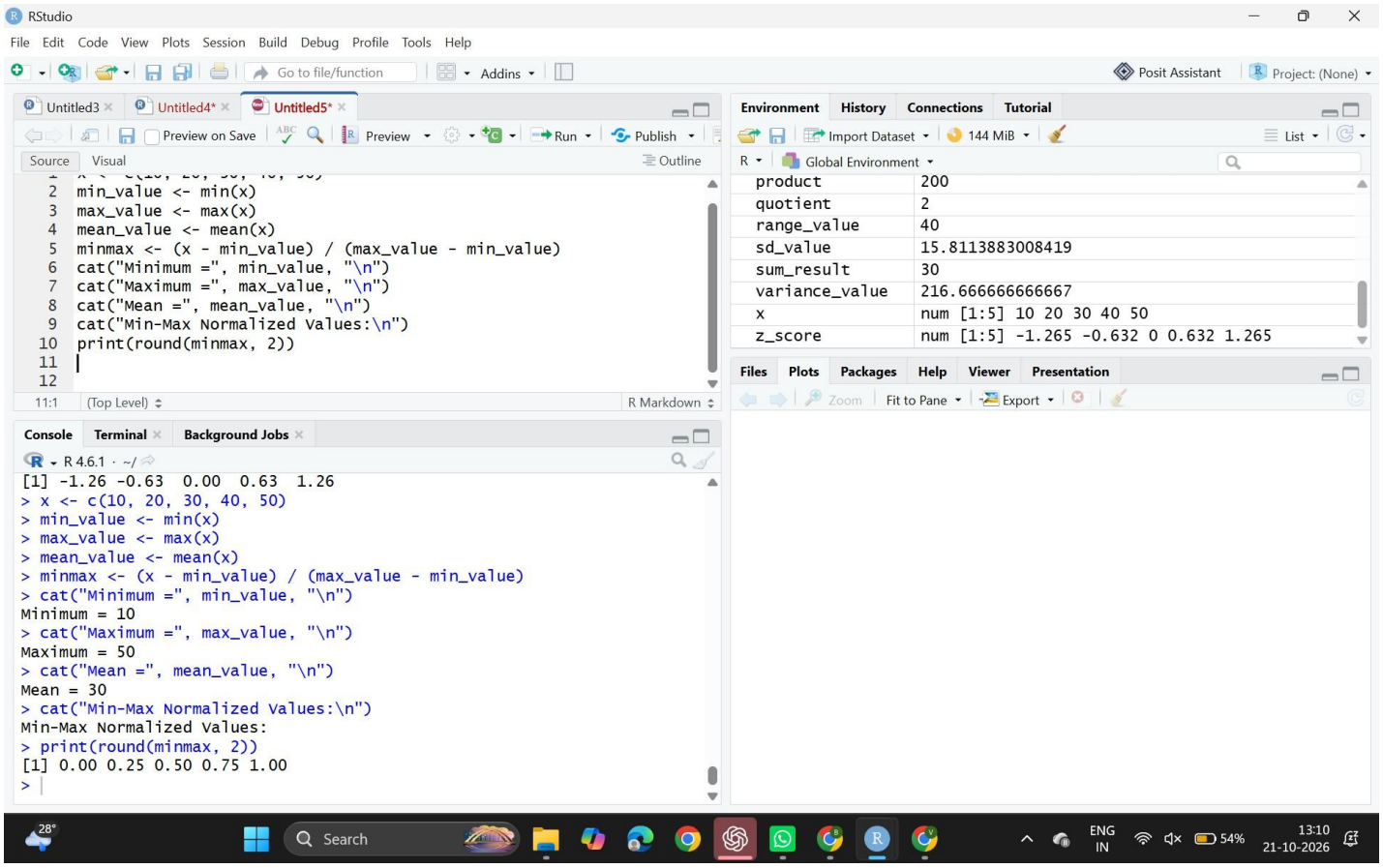
# Experiment-11



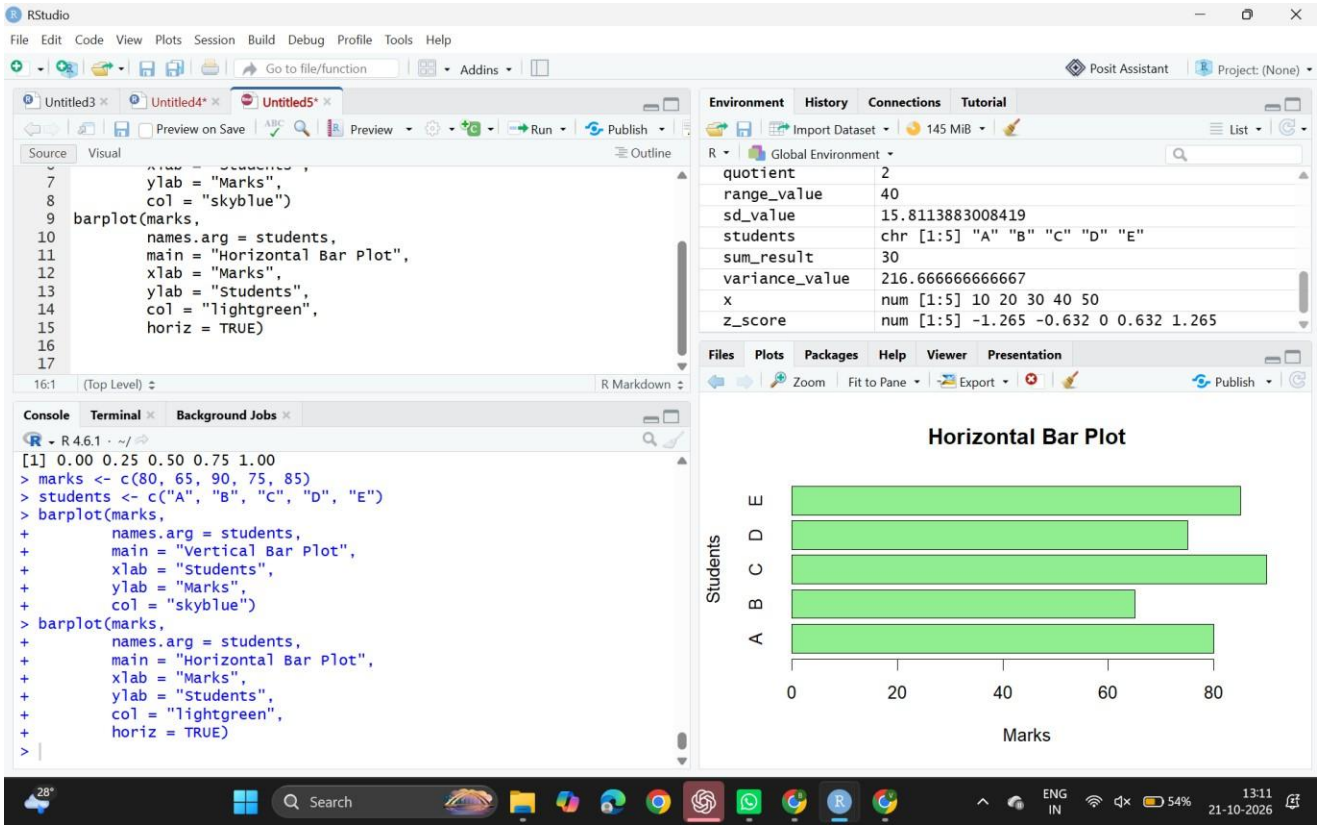

# Experiment-12



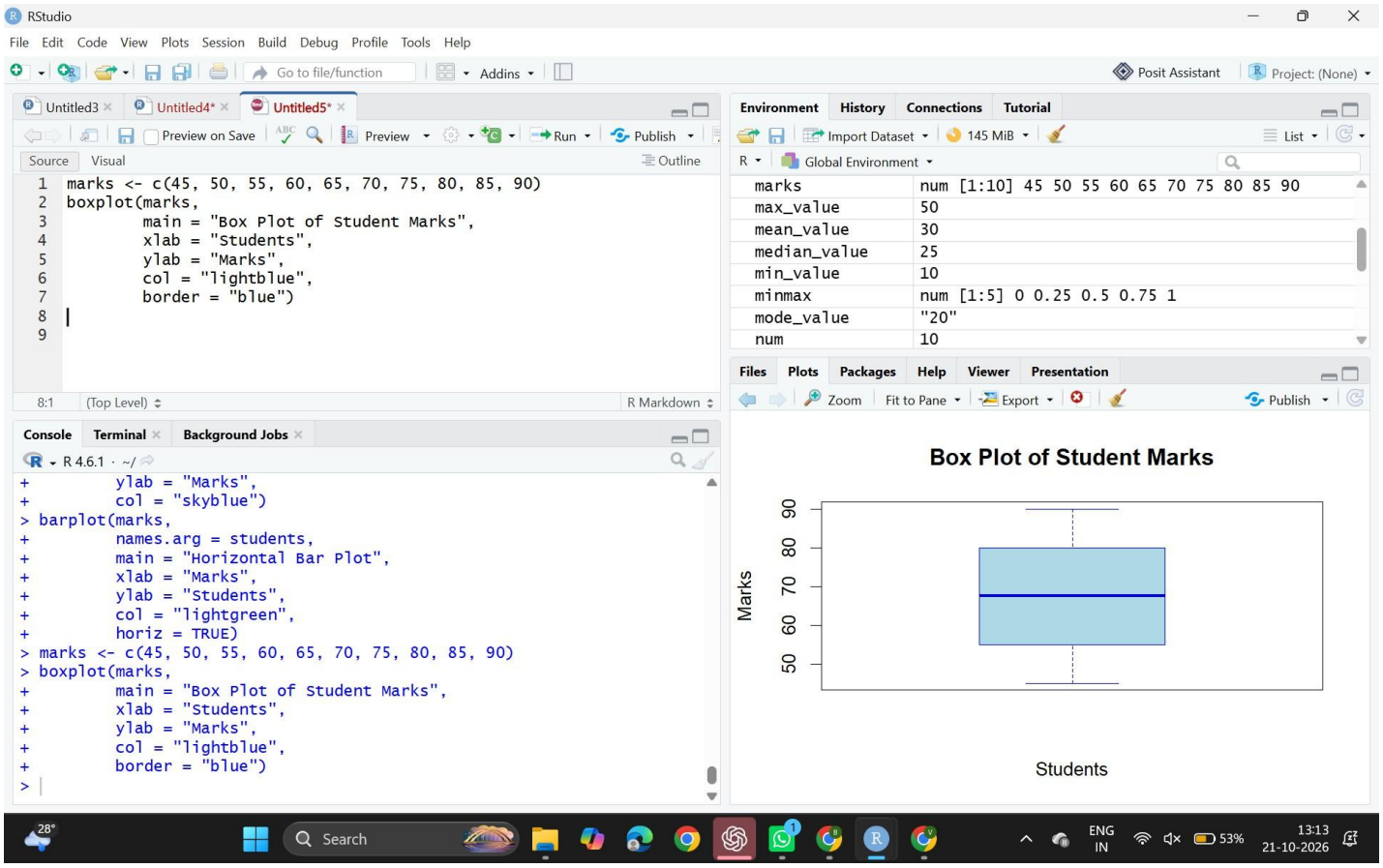
# Experiment-13



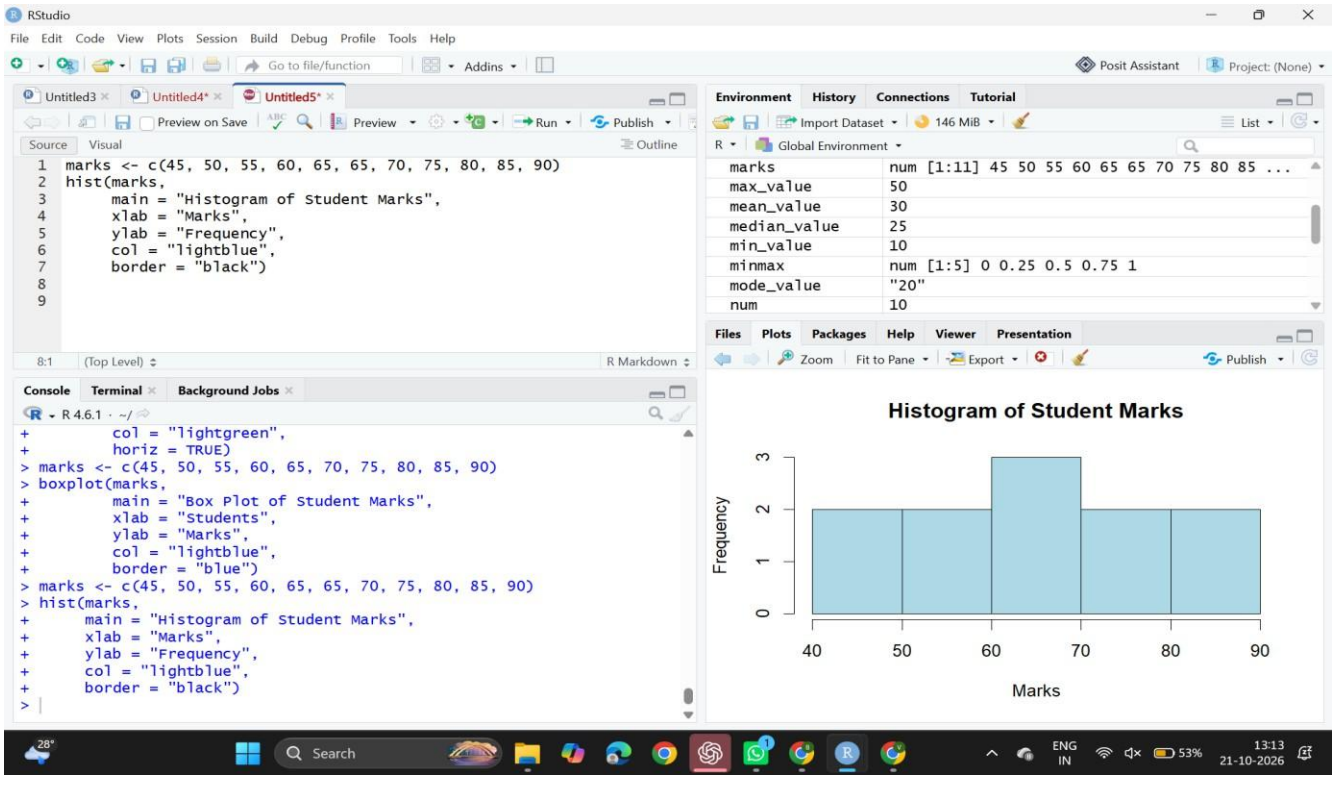

# Experiment-14



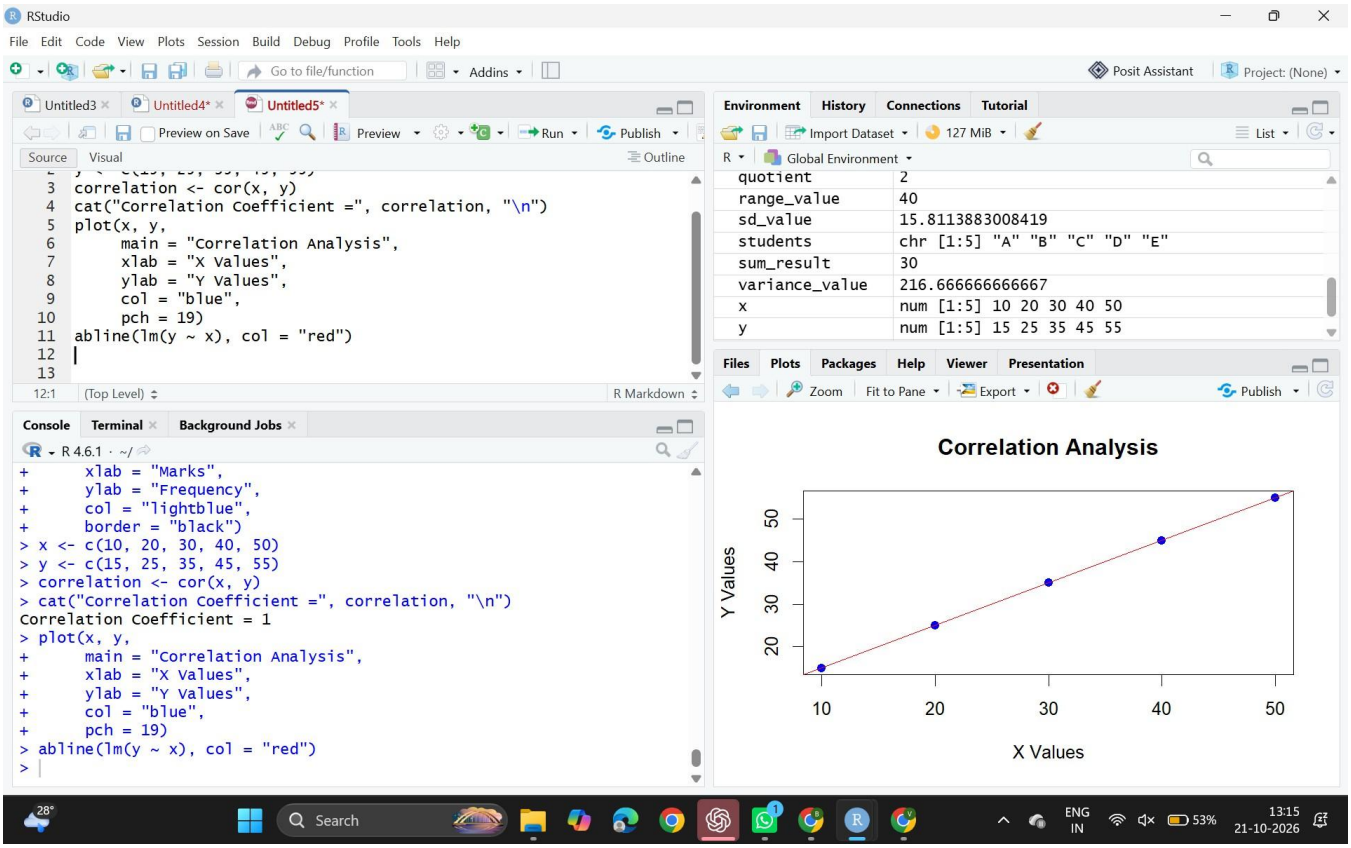
# Experiment-15



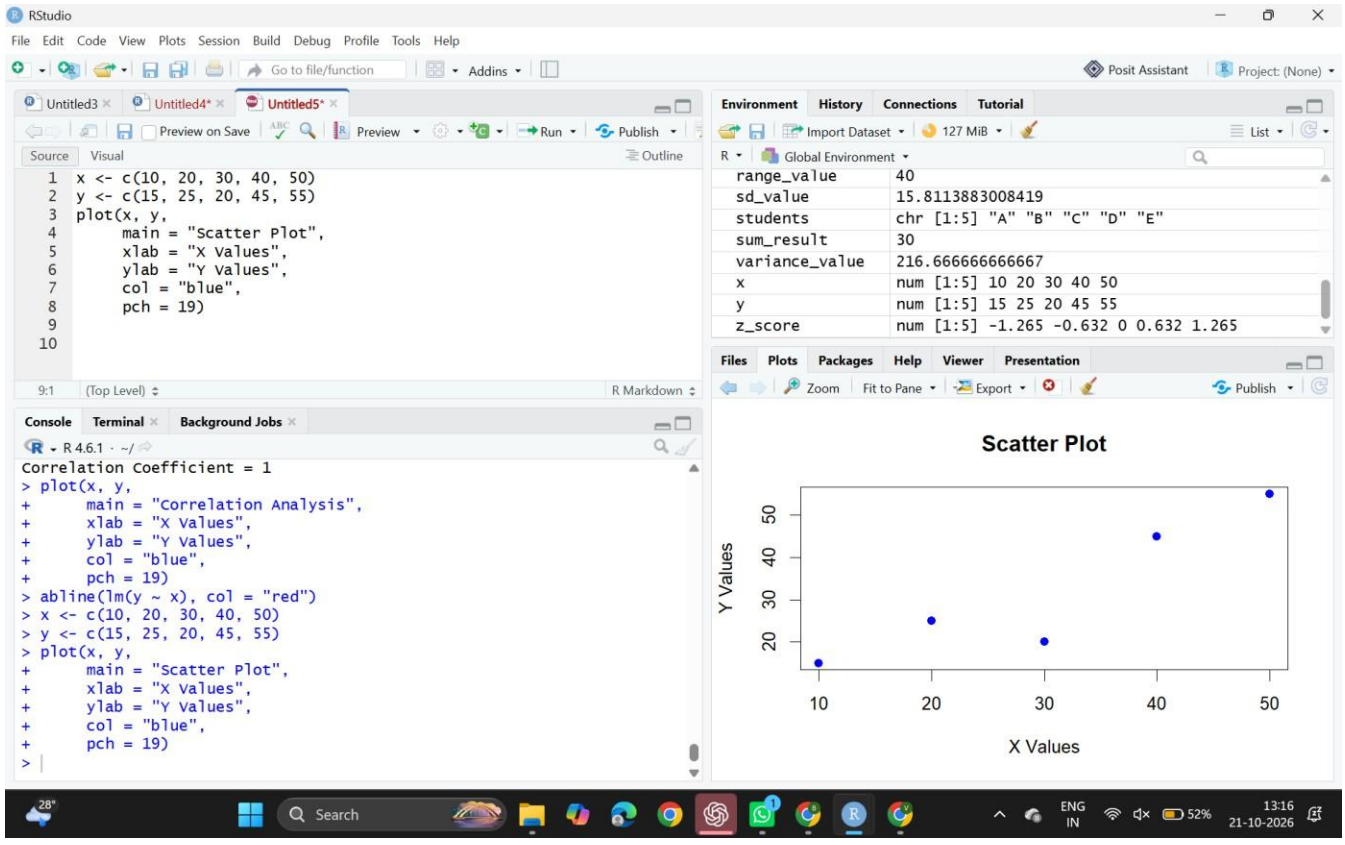

# Experiment-16



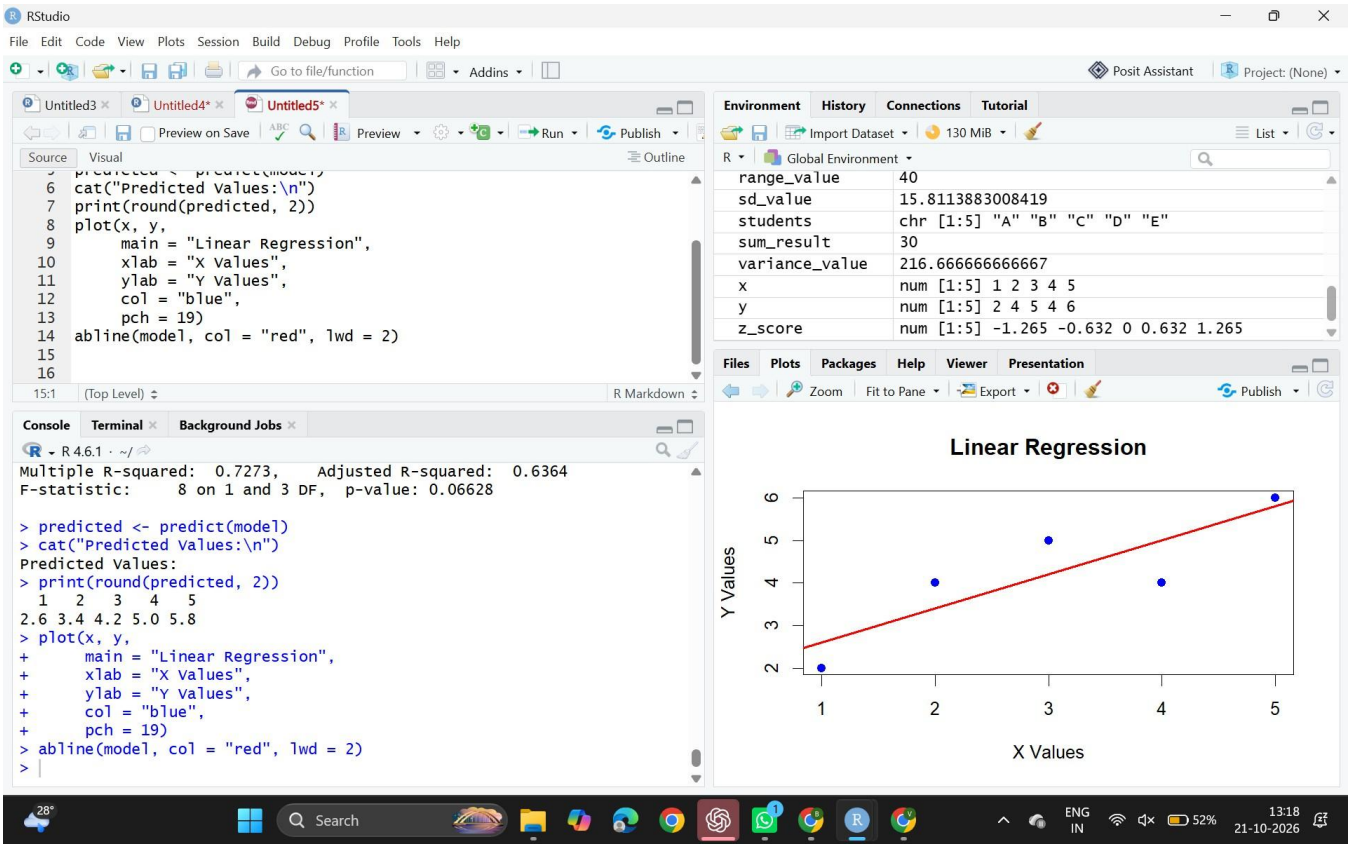
# Experiment-17




# Experiment-18



# Experiment-19




# Experiment-20

